浙江大学实验报告

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课程名称：人工智能　　任课老师：龚小谨

实验名称：利用华为云运行推理任务　　实验日期：2021/6/15

1 实验目的和要求

（分点简要说明本次实验需要进行的工作和最终的目的）

大家以个人为单位完成，在华为云上申请一个装有 atlas310 设备的 ECS 服务器，然后完成环境配置，最后在服务器上运行推理任务的 demo。

主要目的是让大家体验一下从准备资源到开发应用再到部署应用的流程。

进行 atlas 设备应用的开发比较复杂，这里不对大家做强制要求，只要能够从官方 demo 中挑选一个能在设备上运行起来即可。

这里会设置一个小 bonus，可以基于已有的官方 demo，尝试组合几个功能完成一个联合任务，例如人脸识别+卡通化，将照片中的人脸转换成卡通风格。

对于强烈感兴趣，愿意钻研相关开发的同学，欢迎以此设备作为最终大作业的平台，创建深度学习相关的推理应用。如果有原创性强的应用 demo，可以联系 huawei 推送至他们的代码仓库。

2 实验原理

CANN（Compute Architecture for Neural Networks）是华为公司针对 AI 场景推出的异构计算架构，通过提供多层次的编程接口，支持用户快速构建基于昇腾平台的 AI 应用和业务。

视频 https://www.bilibili.com/video/BV1m64y1D7Q1?share_medium=android&share_plat=android&share_source=WEIXIN&share_tag=s_i×tamp=1620612369&unique_k=KugHEJ

https://gitee.com/ascend/samples/tree/master 有一些 demo，我们准备好环境.

Bonus 的话:

可以上色后卡通化

https://gitee.com/ascend/samples/tree/master/python/level2_simple_inference/6_other/colorization_picture

https://gitee.com/ascend/samples/tree/master/cplusplus/contrib/cartoonGAN_picture

3 实验内容

（分点阐述实验步骤）

1. 配置服务器

2. 用 Xshell 登录服务器

3. 替换 Ubuntu 系统的 apt-get 源。

没有创建文件夹

```
HwHiAiUser@ecs-6618:/$ cd home
HwHiAiUser@ecs-6618:/home$ ls
HwHiAiUser@ecs-6618:/home$ mkdir HwHiAiUser
mkdir: cannot create directory ‘HwHiAiUser’: Permission denied
HwHiAiUser@ecs-6618:/home$ ls
HwHiAiUser@ecs-6618:/home$ cd
-bash: cd: /home/HwHiAiUser: No such file or directory
HwHiAiUser@ecs-6618:/home$ cd..
cd..: command not found
```

这个用户怎么什么权限都没有
解决方法： root 给他的目录加 777 ，然后用户就可以在里面 mkdir 了.

在 root 用户下设置 python 的软链接

软链接有问题，没法设置.
bash: /usr/local/python3.7.5/bin/python3.7.5: No such file or directory
apt 显示已安装，但是 python version 不显示
python3 可以显示，
解决方法:那就自己安装一个.（其实是因为不是 AI 增强型）

出现了一些 warning

```
WARNING: The scripts f2py, f2py3 and f2py3.7 are installed in '/home/Hw
HiAiUser/.local/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress
this warning, use --no-warn-script-location.
  WARNING: The script isympy is installed in '/home/HwHiAiUser/.local/b
in' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress
this warning, use --no-warn-script-location.
  WARNING: The script chardetect is installed in '/home/HwHiAiUser/.loc
al/bin' which is not on PATH.
  Consider adding this directory to PATH or, if you prefer to suppress
this warning, use --no-warn-script-location.
```

```
Successfully installed attrs-21.2.0 certifi-2021.5.30 cffi-
1.12.3 chardet-4.0.0 decorator-5.0.9 idna-2.10 mpmath-1.2.1 numpy-
1.17.2 pathlib2-2.3.5 protobuf-3.17.3 psutil-5.8.0 pycparser-
2.20 pyyaml-5.4.1 requests-2.25.1 scipy-1.6.3 six-1.16.0 sympy-
1.4 urllib3-1.26.5 xlrd-1.2.0
WARNING: You are using pip version 19.2.3, however version 21.1.2 is av
ailable.
You should consider upgrading via the 'pip install --
upgrade pip' command.
```

不管他

4. 安装 CANN 开发环境源

安装硬件驱动

硬件驱动 A300 下载，我遇到了 403，也可能 400.
解决方法：Root 可以传上,但是用户 scp permission denied.
root 上传上去找不到
scp .\A300-3010-npu-driver_20.2.0_ubuntu18.04-x86_64.run root@ip:home 会把文件重命名为 home.
解决方法： ip:/home/HwHiAiUser/downloads 这样才是正确的路径 前面要/home
别的都可以正常下载.
Linux 下调用动态库和 windows 不一样.linux 可执行程序是靠配置文件去读取路径的,因此有些时候需要设置路径
具体操作如下

export LD_LIBRARY_PATH=/home/.....(动态库的目录)

注意，与 DOS/Window 不同，UNIX 类系统环境变量中路径名用冒号分隔，不是分号。另外，软件越装越多，环境变量越添越多，为了避免造成混乱，建议所有语句都添加在文件结尾，按软件的安装顺序添加。
$PATH：决定了 shell 将到哪些目录中寻找命令或程序
${HOME}：首先， HOME 是个变量，它代表你的 home 目录，变量前必须加上 $ 符号，否则会报错 **export $PATH="$PATH:路径 1:路径 2:...:路径 n" （或"PATH=$PATH:路径 1:路径 2:...:路径 n" ）**，意思是可执行文件的路径包括原先设定的路径，也包括从"路径 1"到"路径 n"的所有路径。当用户输入一个一串字符并按回车后，shell 会依次在这些路径里找对应的可执行文件并交给系统核心执行。那个"$PATH"表示原先设定的路径仍然有效，注意不要漏掉。

5. 替换 pip 源

Toolkit 装完毕

6.安装 mindstudio

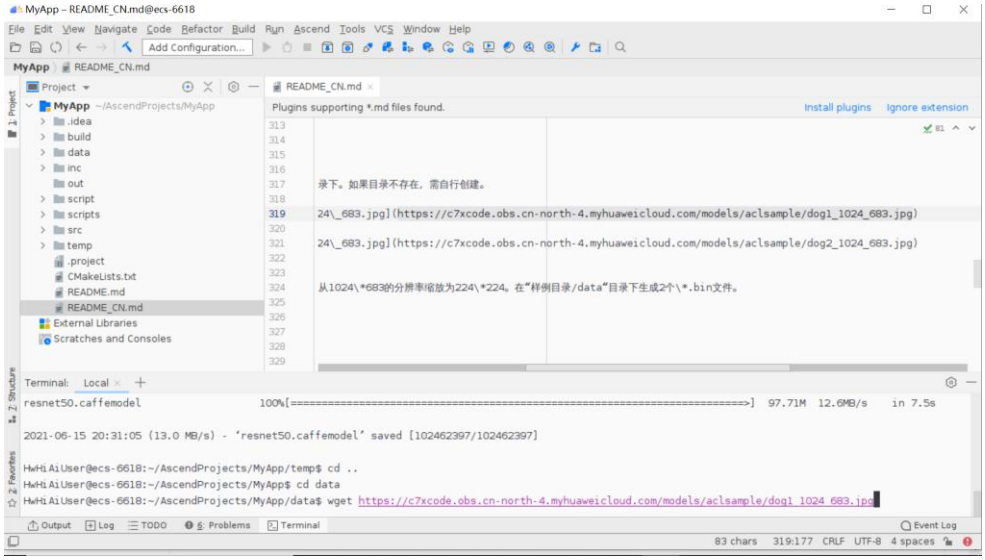
这个 shell 非常智能 会检查依赖然后告诉你缺乏哪些依赖.
XSHELL 不能在中国买，显示 x11 要收费. 可能还得装 mobaxterm
: error while loading shared libraries: libpython3.7m.so.1.0: cannot open shared object file: No such file or directory
解决方法: `cp /usr/local/python3/lib/libpython3.6m.so.1.0 /usr/lib/`

自己安装 jdk

提示没有指定 java 环境变量，主要原因是没有安装 jdk，因此进入 root 用 户，安装 jdk：有了，不过确实延迟比较大.

7.尝试运行工程

视频 53 分钟



[ERROR] RUNTIME(5109,main):2021-06-15-21:47:01.293.558 [runtime.cc:626]5109 InitSocType:Call halGetDeviceInfo failed: drvRetCode=2, module type=0, info type=1.
[ERROR] RUNTIME(5109,main):2021-06-15-21:47:01.293.610 [runtime.cc:704]5109 Init:Init SocType failed.
[ERROR] RUNTIME(5109,main):2021-06-15-21:47:01.293.641 [api_c.cc:297]5109 Instance:Runtime::Instance == nullptr
[ERROR] RUNTIME(5109,main):2021-06-15-21:47:01.293.673 [api_c.cc:1010]5109 rtDeviceReset:api is NULL!
[ERROR] RUNTIME(5109,main):2021-06-15-21:47:01.293.698 [api_c.cc:1010]5109 rtDeviceReset:Null api pointer, ErrCode=507000
[ERROR] ASCENDCL(5109,main):2021-06-15-21:47:01.293.726 [device.cpp:113]5109

aclrtResetDevice: reset device 0 failed, runtime result = 507000.
[ERROR] reset device 0 failed, errorCode = 507000

```
/home/HwHiAiUser/AscendProjects/MyApp/src
Information: Execution command: export LANG=C && export DDK_PATH=/hom
             NPU_HOST_LIB=/home/HwHiAiUser/Ascend/ascend-toolkit/3.3.0.a
             /home/HwHiAiUser/AscendProjects/MyApp/build/intermediates &&
             -Dtarget=SOC -DCMAKE_BUILD_TYPE=Release -DCMAKE_CXX_COM
Information: -- env INC_PATH: /home/HwHiAiUser/Ascend/ascend-toolkit/3.3.0.
Information: -- env LIB_PATH: /home/HwHiAiUser/Ascend/ascend-toolkit/3.3.0.a
Information: -- Configuring done
Information: -- Generating done
Information: -- Build files have been written to: /home/HwHiAiUser/AscendProj
Information: [100%] Built target main
```

Build 没有 successful
解决：尝试命令行，我不知道 cann-toolkit 装哪里了

1. 开发环境上，设置模型转换依赖的环境变量。

   ${install_path}表示开发套件包Ascend-cann-toolkit所在的路径。

```
export
PATH=${install_path}/atc/ccec_compiler/bin:${install_path}/atc/bin:
$PATH
export ASCEND_OPP_PATH=${install_path}/opp
```

2. 开发环境上，设置环境变量，编译脚本src/CMakeLists.txt通过环境变量所设置的头文件、库文件的路径来编译代码。

   如下为设置环境变量的示例，请将$HOME/Ascend/ascend-toolkit/latest/*{os_arch}*替换为开发套件包Ascend-cann-toolkit下对应架构的FwkACLlib的路径。

   - 当运行环境操作系统架构为x86时，执行以下命令：

好像找到了/home/HwHiAiUser/Ascend/ascend-toolkit/3.3.0.alpha006
Path 可以分两段写吗,好像可以的.
1.修改/etc/profile 文件 所有用户的 shell 都有权使用这些环境变量
这些好像都是开发环境，运行不需要这些.
问题 ：
compilation terminated.
CMakeFiles/main.dir/build.make:62: recipe for target 'CMakeFiles/main.dir/utils.cpp.o' failed
make[2]: *** [CMakeFiles/main.dir/utils.cpp.o] Error 1
CMakeFiles/Makefile2:67: recipe for target 'CMakeFiles/main.dir/all' failed
make[1]: *** [CMakeFiles/main.dir/all] Error 2
Makefile:129: recipe for target 'all' failed
make: *** [all] Error 2
缺 acl

```
#用于设置python3.7.5库文件路径
export LD_LIBRARY_PATH=/home/HwHiAiUser/Ascend/nnrt/latest/acllib/lib64:/home/HwHiAiUser/Ascend/ascend-toolki
t/latest/acllib/lib64:/usr/local/python3.7.5/lib:$LD_LIBRARY_PATH #如果用户环境存在多个python3版本，则指定使>
用python3.7.5版本
export PATH=$JAVA_HOME/bin:/usr/local/python3.7.5/bin:$PATH
export PYTHONPATH=/home/HwHiAiUser/Ascend/nnrt/latest/pyACL/python/site-packages/acl:/home/HwHiAiUser/Ascend/
ascend-toolkit/latest/pyACL/python/site-packages/acl:/home/HwHiAiUser/Ascend/ascend-toolkit/latest/toolkit/py
thon/site-packages:${PYTHONPATH}
export ASCEND_AICPU_PATH=/home/HwHiAiUser/Ascend/nnrt/latest:/home/HwHiAiUser/Ascend/ascend-toolkit/latest:${
ASCEND_AICPU_PATH}
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export PATH=/home/HwHiAiUser/Ascend/ascend-toolkit/3.3.0.alpha006/atc/ccec_compiler/bin:/home/HwHiAiUser/Asce
nd/ascend-toolkit/3.3.0.alpha006/atc/bin:$PATH
export ASCEND_OPP_PATH=/home/HwHiAiUser/Ascend/ascend-toolkit/3.3.0.alpha006/opp:/home/HwHiAiUser/Ascend/asce
nd-toolkit/latest/opp:
export TOOLCHAIN_HOME=/home/HwHiAiUser/Ascend/ascend-toolkit/latest/toolkit:

~
~
```

应该是环境变量错了，不知道该怎么改

好奇怪为啥有的自己看有,但是 find 不到. find 不常用，因为速度慢，耗费硬盘空间。通常我们先使用 whereis 或者 locate 来检查,如果 找不到,才用 find 搜索。为什么？因为 whereis 与 locate 是使用数据库来搜索数据，所以相当快，而且并没有实际搜索硬盘，比较省时。

重新 build model 看看

```
Output   Detail
021-06-16 14:24:29  Start to convert model
021-06-16 14:24:29  export PATH=/usr/local/python3.7.5/bin:$PATH:/home/HwHiAiUser/Ascend/ascend-too
021-06-16 14:24:51  call drvMngGetConsoleLogLevel failed , g_conLogLevel = 3
021-06-16 14:24:51  ATC start working now, please wait for a moment.
```

./main: error while loading shared libraries: libascendcl.so: cannot open shared object file: No such file or directory

不知道为什么会缺依赖. 找到了 so 后还是会出错.

我试了试 npu-smi info

call drvMngGetConsoleLogLevel failed , g_conLogLevel = 3

Failed to get card number. 这是因为用的不是 310

这个 310 到底是什么机制?

```
export LD_LIBRARY_PATH=${install_path}/atc/lib64
```

我很怀疑这样行不行

转换模型的时候

```
/model$ atc --output_type=FP32 --input_shape="input_1:1,224,224,3"  --
input_format=NHWC --output="vgg16_cat_dog" --soc_version=Ascend310 --
insert_op_conf=insert_op.cfg --framework=3 --model="./vgg16_cat_dog.pb"

call drvMngGetConsoleLogLevel failed , g_conLogLevel = 3
ATC start working now, please wait for a moment.
ATC run failed, Please check the detail log, Try 'atc --help' for more
information
```

应该是驱动有问题.

```
E19024: The env ASCEND_OPP_PATH[/home/HwHiAiUser/Ascend/ascend-
toolkit/3.3.0.alpha006/opp:/home/HwHiAiUser/Ascend/ascend-
toolkit/latest/opp:] is invalid when loading the tiling lib
```

这个应该是不用加冒号

原来是因为没有买 AI 加速型.

8.重新配置

--no-check-certificate
除了文档的环境外，还有 python3 的环境
https://gitee.com/ascend/samples/blob/master/python/environment/python_ENV/README_300_CN.md
需要配置
不知道是不是对的，因为他说 python3.6,而我是 3.7
ModuleNotFoundError: No module named 'cv2'
怎么知道 root 把 opencv 装再哪里?
pip show opencv
尝试 1：可以把位置放在 bashrc 中

```
ollecting opencv-python
 Downloading https://pypi.tuna.tsinghua.edu.cn/packages/bb/08/9dbc183a3ac6baa95fabf749ddb531bd26256edfff5b6c
195eca26258e9/opencv-python-4.5.1.48.tar.gz (88.3MB)
   100% |[redacted]| 88.3MB 6.2kB/s
   Complete output from command python setup.py egg_info:
   Traceback (most recent call last):
     File "<string>", line 20, in <module>
     File "/tmp/pip-build-3f4sxB/opencv-python/setup.py", line 10, in <module>
       import skbuild
   ImportError: No module named skbuild

   ----------------------------------------
ommand "python setup.py egg_info" failed with error code 1 in /tmp/pip-build-3f4sxB/opencv-python
ou are using pip version 7.1.2, however version 21.1.2 is available.
ou should consider upgrading via the 'pip install --upgrade pip' command.
wHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_picture/src$
```

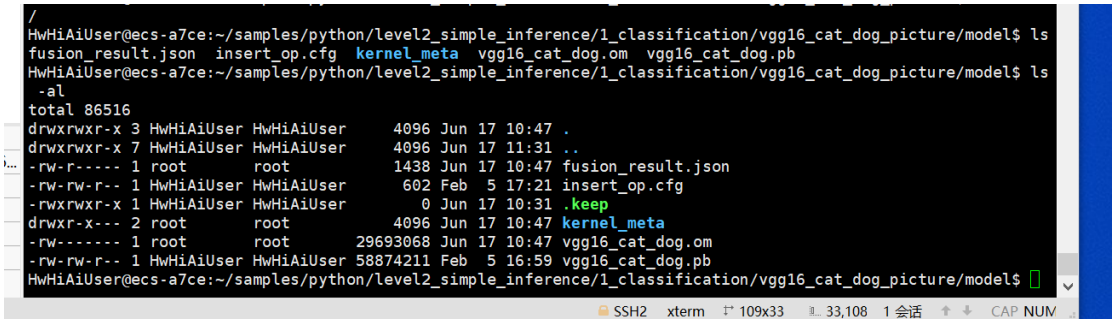
一个是可以 装个低版本的 opencv，3.4.x 的
一个是要升级 pip
好多错误，基本上说无权限，
是不是不应该是 2.7 ． 而是 3.7
pip -V
pip 7.1.2 from /usr/local/lib/python2.7/dist-packages/pip-7.1.2-py2.7.egg (python 2.7)
Building wheel for opencv-python (PEP 517) ... /
卡住加上版本信，采用低版本的 opencv。`pip install opencv-python==3.4.9.31`

然后加了离线推理环境

```
#如果用户环境存在多个python3版本，则指定使用python3.7.5版本
export PATH=/usr/local/python3.7.5/bin:$PATH
export PYTHONPATH=/home/HwHiAiUser/Ascend/ascend-toolkit/latest/pyACL/python/s
r/Ascend/ascendtoolkit/latest/toolkit/python/site-packages:/home/HwHiAiUser/As
ite-packages/acl:{$PYTHONPATH}
export ASCEND_AICPU_PATH=/home/HwHiAiUser/Ascend/nnrt/latest:/home/HwHiAiUser/
ASCEND_AICPU_PATH}

export PYTHONPATH=$HOME/samples/python/common/:$PYTHONPATH
export ASCEND_OPP_PATH=/home/HwHiAiUser/Ascend/ascend-toolkit/latest/opp:
-- INSERT --
```

还是不行



```
/
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_picture/model$ ls
fusion_result.json  insert_op.cfg  kernel_meta  vgg16_cat_dog.om  vgg16_cat_dog.pb
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_picture/model$ ls
 -al
total 86516
drwxrwxr-x 3 HwHiAiUser HwHiAiUser     4096 Jun 17 10:47 .
drwxrwxr-x 7 HwHiAiUser HwHiAiUser     4096 Jun 17 11:31 ..
-rw-r----- 1 root       root           1438 Jun 17 10:47 fusion_result.json
-rw-rw-r-- 1 HwHiAiUser HwHiAiUser      602 Feb  5 17:21 insert_op.cfg
-rwxrwxr-x 1 HwHiAiUser HwHiAiUser        0 Jun 17 10:31 .keep
drwxr-x--- 2 root       root           4096 Jun 17 10:47 kernel_meta
-rw------- 1 root       root       29693068 Jun 17 10:47 vgg16_cat_dog.om
-rw-rw-r-- 1 HwHiAiUser HwHiAiUser 58874211 Feb  5 16:59 vgg16_cat_dog.pb
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_picture/model$
```

把 om 模型权限改掉,成功了

4bonus：

首先，把黑白的图转化为彩色图.　然后识别.

samples: CANN Samples – Gitee

遇到的问题:
1.没有 atc,还是环境变量不对这个是离线推理也就是运行环境,
2.atc 是开发环境，也就是 Model transformation and operator develop 这是模型转换的
3.path 分两行写可以吗
可以
4. The env ASCEND_OPP_PATH[/home/HwHiAiUser/Ascend/ascend-toolkit/latest/opp:] is invalid when loading the tiling lib
因为多了一个冒号，最后不用加冒号也不用加$ascendopp

5. ModuleNotFoundError: No module named 'PIL'
不要用 python3.6
命令 : python3 colorize.py ../data/ 即可

6. AttributeError: 'NoneType' object has no attribute 'width'
好像 image 没有读进来
把图片改成一样大就可以了.

5 实验结果和分析

（使用图片和文字叙述实验结果，并对这些结果进行适当分析）

```
        LD_LIBRARY_PATH includes :
                /home/HwHiAiUser/Ascend/ascend-toolkit/latest/atc/lib64:
        PYTHONPATH includes :
                /home/HwHiAiUser/Ascend/ascend-toolkit/latest/atc/python/site-packages:
                /home/HwHiAiUser/Ascend/ascend-toolkit/latest/toolkit/python/site-packages:
        ASCEND_OPP_PATH includes :
                /home/HwHiAiUser/Ascend/ascend-toolkit/latest/opp:
        TOOLCHAIN_HOME includes :
                /home/HwHiAiUser/Ascend/ascend-toolkit/latest/toolkit:
[Toolkit] [20210615-19:48:34] [INFO] If your service is started using the shell script, you can call the /hom
e/HwHiAiUser/Ascend/ascend-toolkit/set_env.sh script to configure environment variables. Note that this scrip
t can not be executed mannually.
[Toolkit] [20210615-19:48:34] [INFO] Ascend-cann-toolkit_3.3.0.alpha006_linux-x86_64.run install success,The
install path is /home/HwHiAiUser/Ascend !
HwHiAiUser@ecs-6618:~/downloads$
```

Toolkit 装完毕

延迟比较大

```
2021-06-15 20:59:56  Convert model command:
2021-06-15 20:59:56  /home/HwHiAiUser/Ascend/ascend-toolkit/3.3
2021-06-15 20:59:56  Model converted successfully.
2021-06-15 20:59:56  Model input path:/home/HwHiAiUser/AscendPr
2021-06-15 20:59:56  Model output path:/home/HwHiAiUser/modelzo
2021-06-15 20:59:56  Model conversion log file path:/home/HwHiA
2021-06-15 20:59:56  Model conversion config file path:/home/Hw
```

预处理成功

终于成功推理了

```
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_pict
 ..
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_pict
data  model  outputs  README_CN.md  scripts  src
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_pict
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_pict
on3 main.py
init resource stage:
Init resource success
Init model resource start...
[Model] create model output dataset:
malloc output 0, size 8
Create model output dataset success
Init model resource success
=== dog.jpg===
in pre_process, use time:0.007152557373046875
in inference, use time:0.003950834274291992
in post_process, use time:0.06495332717895508
=== cat.jpg===
in pre_process, use time:0.002527952194213867
in inference, use time:0.003930568695068359
in post_process, use time:0.011335611343383789
acl resource release all resource
dvpp resource release success
Model release source success
acl resource release stream
acl resource release context
Reset acl device  0
Release acl resource success
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_pict
```

```
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/6_other/colorization_picture/src$ python3 colori
ze.py ../data/
init resource stage:
Init resource success
Init model resource start...
[Model] create model output dataset:
malloc output 0, size 25088
Create model output dataset success
Init model resource success
Execute end
acl resource release all resource
Model release source success
acl resource release stream
acl resource release context
Reset acl device  0
Release acl resource success
```

推理成功

输出如下图

推理成功！

原图

之前我们可以识别这是狗

推理成功！

```
..
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_picture$ cd
HwHiAiUser@ecs-a7ce:~/samples/python/level2_simple_inference/1_classification/vgg16_cat_dog_picture/src$
on3 main.py
init resource stage:
Init resource success
Init model resource start...
[Model] create model output dataset:
malloc output 0, size 8
Create model output dataset success
Init model resource success
=== dog2.jpg===
in pre_process, use time:0.0030875205993652344
in inference, use time:0.003970146179199219
in post_process, use time:0.014133453369140625
=== cat.jpg===
in pre_process, use time:0.0025377273559570312
in inference, use time:0.003929853439331055
in post_process, use time:0.011345624923706055
acl resource release all resource
dvpp resource release success
Model release source success
acl resource release stream
acl resource release context
Reset acl device  0
Release acl resource success
```

接下来是识别刚刚产生的彩色照片.

遇到了 resize error,是大小的问题, 把大小调整和猫一样就可以了.

如下图, 可以看到红色 dog 推理.

Dog 字好小 ,不过可以认出来这是 dog

修改一下代码,如果是猫就不画圆弧

```
    font = ImageFont.load_default()
    if pre_index==1:
        draw.arc((0, 0,100,100) , start=0, end=300, fill='red',width=3)
    draw.text((10, 50),"this is a "+ CLS[pre_index], font=font, fill=255)
    origin_img.save(output_path)


main():
"""
```

说明确实是狗

bonus 组合完毕，首先黑白变彩色，然后判断.